Examen final de Xarxes de Computadors (XC)	Grau en Ingeniería Informàtica	8/6/2026	Primavera 2026

Duration: 2h45m. The questions have a single correct answer. They are scored p, $-p/3$ if wrong, 0 for no answer. Mark the answers on the back.

QUIZ (2,5 points)

Which OSI layer is responsible for

1. Routing datagrams between different networks: A. Physical. B. Data Link. C. Network. D. Transport.
2. Reliable data transfer between applications: A. Network. B. Application. C. Data Link. D. Transport.

Which is the maximum number of IP addresses that can be assigned to hosts in the subnet 192.168.10.0/27?

3. A. 32 B. 62 C. 30 D. 16

Which is the broadcast address of the subnet 172.16.8.0/21?

4. A. 172.16.15.0 B. 172.16.16.255 C. 172.16.8.255 D. 172.16.15.255

Which is a valid subnet address?

5. A. 10.20.64.0/18 B. 10.20.63.0/19 C. 10.20.64.16/20 D. 10.20.127.0/18

A 4000-byte datagram is fragmented at a router with a Maximum Transmission Unit (MTU) of 1500 bytes. What is the offset field value in the datagram header (in 8-byte words) of the third fragment?

6. A. 370 B. 148 C. 185 D. 160

Which kind of ICMP message does the *traceroute* command produce to map the route a packet takes to a destination?

7. A. ICMP type 3 (Destination unreachable) B. ICMP type 5 (Redirect Message) C. ICMP type 8 (Echo Message) D. ICMP type 11 (Time Exceeded)

When a host performs an ARP request, what type of frame does it send out, and what type of frame does the target reply with?

8. A. Broadcast request, unicast reply B. Unicast request, unicast reply C. Unicast request, broadcast reply D. Broadcast request, broadcast reply

What is the JAM signal used for in CSMA/CD?

9. A. It marks the end of the frame. B. It requests the receiver to discard the frame. C. It resets the backoff counter on all stations. D. It produces an error CRC so all stations detect the collision.

What mechanism or protocol is used to

10. Solve broadcast storms caused by L2 loops: A. Gratuitous ARP B. ICMP C. STP D. Virtual Private Networks (VPNs)
11. Allow multiple devices in a private network to share a single public IPv4 address when accessing the Internet: A. VLANs B. NAT C. IP D. DHCP
12. Reliably transfer a web page from a server to a client, ensuring ordered delivery and retransmission of lost data: A. UDP B. HTTP C. IP D. TCP
13. Resolve a domain name into an IP address: A. IP B. DHCP C. ARP D. DNS

Which sequence of segments correctly describes TCP's connection establishment (three-way handshaking)?

14. A. ACK -> SYN -> SYN+ACK B. SYN -> SYN+ACK -> ACK C. SYN -> ACK -> SYN+ACK D. SYN -> ACK -> SYN -> ACK

What is the purpose of the TCP advertised window (awnd)?

15. A. To indicate the maximum segment seize negotiated during connection setup. B. To inform the sender of the free space in the receiver's buffer for flow control. C. To control network congestion by limiting the sender's transmission rate. D. To track the number of unacknowledged segments in the network.

Which DNS Resource Record (RR) type is used to represent...

16. An IPv6: A. IP B. CNAME C. A D. AAAA
17. A nameserver: A. NS B. A C. MX D. CNAME
18. An alias: A. CNAME B. A C. NS D. ALIAS

Which MIME header defines the type and subtype of content?

19. A. Content-Transfer-Encoding B. Content-Type C. MIME-Version D. Charset

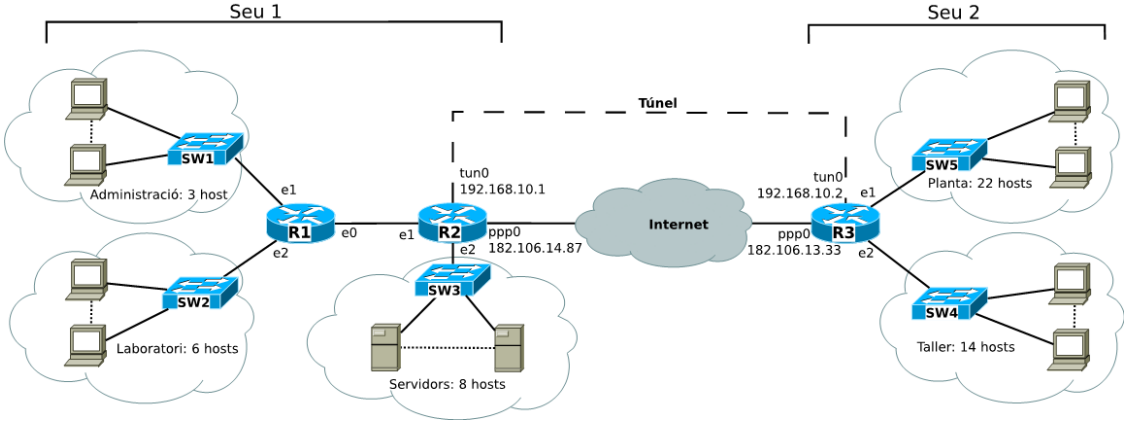
Examen final. Xarxes de Computadors (XC), Grau en Enginyeria Informàtica	08/06/2026	Spring 2026	1

Total time: 2 hours 45 minutes. Single answer. All score p if correct, -p/3 if wrong, 0 if no answer. Clearly mark answers in pencil on the back.

IP Problem IP (2.5 points; approximate time 40 minutes)

The figure shows an organisation's network and its connection to the Internet. Site 1 is the main site and Site 2 is the production plant. There are five subnets: administration (A), laboratory (L), servers (S), plant (P), and workshop (T). The figure indicates how many devices are in each one. The subnets are Ethernet at 100 Mbps. The Internet Service Providers (ISPs) assign the public IP address to each site: 182.106.14.87/26 (ISP1) and 182.106.13.33/26 (ISP2) respectively. The DNS service is external and provided by an Internet Service Provider (ISP).

Both sites are connected via an IP tunnel. All traffic from Site 2 is routed through the tunnel and accesses the Internet through the tunnel and R2.



If the access router of each ISP has the first IP address of the corresponding range, the IP addresses of GW1 and GW2 are:

1.	GW1	A	182.106.14.1	B	182.106.14.65	C	182.106.14.129	D	182.106.14.33
2.	GW2	A	182.106.13.65	B	182.106.13.33	C	182.106.13.129	D	182.106.13.1

Private addressing 192.168.144.0/24 is used to assign IP addresses to subnets A, L, S, P, T, adjusting the network prefix so that the number of unassigned addresses within each subnet is minimal (adjusting the mask size to the number of devices in each subnet).

The mask of each subnet (A, L, S, P, T) is:

3.	A	/29 /28 /28 /27 /27	B	/28 /28 /28 /26 /26	C	/28 /28 /28 /28 /27	D	/28 /27 /27 /26 /26

If the R1-R2 link is assigned the subnet 192.168.144.0/30 and the first subnet that follows is A, its network prefix will be:

4.	A	192.168.144.8/29	B	192.168.144.4/29	C	192.168.144.4/28	D	192.168.144.8/28

The routing table of R2 (prefix, gw, if), where prefix is A, L, S, P, T, R1-R2 (192.168.144.0/30), tunnel (192.168.10.0/30), 182.106.14.87/26 (ISP1), 182.106.13.33/26 (ISP2); and gw is GW1, GW2, R.interface.

5. A (R1-R2, - , e1) (S, - , e2) (GW1/32, - , ppp0) (tunnel, - , tun0) (A+L, R1.e0, e1) (P+T, 192.168.10.2, R3.tun0) (ISP2, GW1, ppp0) (0.0.0.0/0, GW1, ppp0)
   B (R1-R2, - , e1) (S, - , e2) (GW1/32, - , ppp0) (tunnel, - , tun0) (A+L, R1.e0, e1) (P+T, 192.168.10.2, tun0) (ISP2, GW1, ppp0) (0.0.0.0/0, GW1, tun0)
   C (R1-R2, - , e1) (S, - , e2) (GW1/32, - , ppp0) (tunnel, - , tun0) (A+L, R1.e0, e1) (P+T, 192.168.10.2, tun0) (ISP2, GW1, ppp0) (0.0.0.0/0, GW1, ppp0)
   D (R1-R2, - , e1) (S, - , e2) (GW1/32, - , ppp0) (tunnel, - , tun0) (A+L, R1.e0, e1) (P+T, R3.ppp0, tun0) (0.0.0.0/0, GW1, ppp0)

To test the configuration, from a device A1 (network A) the ping command is run to different destinations. Indicate the fields of the IP datagram header (source IP, destination IP, and Protocol) for each destination and interface (for ICMP ECHO RQ message only).

A1> *ping* R2.ppp0

6.	R1e1	A	A1,R2ppp0,ICMP	B	R1e1,R2ppp0,ICMP	C	SW1,R2ppp0,ICMP	D	A1, R2ppp0,UDP

A1> *ping* P1 (plant device)

7.	R3e1	A	R1e1,R3e1,ICMP	B	R2ppp0,R3ppp0,IP	C	A1,P1,ICMP	D	R2ppp0,P1,ICMP
8.	R2ppp0	A	A1,P1,ICMP	B	R2ppp0,R3ppp0,IPinIP	C	192.168.10.1,P1,IPinIP	D	R2ppp0,R3e1,ICMP

A1> *ping* DNS (external DNS server)

9.	R1e0	A	A1,DNS,ICMP	B	A1,DNS,DNS	C	R1e1,DNS,ICMP	D	R1e0,DNS,IPinIP
10.	R2ppp0	A	A1,DNS,ICMP	B	R1e0,R3ppp0,ICMP	C	192.168.10.1,DNS,IPinIP	D	R2ppp0,DNS,ICMP

To test the configuration, from a device A1 (network A) the traceroute command is run to different destinations. Indicate the list of IP addresses returned.

A1> *traceroute* P1

11.	A	R1e1,R2e1,GW1,GW2,R3ppp0,P1	B	R1e1,R2e1,R3ppp0,P1	C	R1e1,R1e2,R2e1,R3e1,P1	D	R1e1,R2e1,192.168.10.1,P1

A1> *traceroute* DNS

12.	A	R1e1,R2e1,R2ppp0, ... DNS	B	R2e1,GW1,GW2,R3ppp0	C	R1e1,R2e1,GW1 ... DNS	D	R1e1,GW1 ... DNS

Within the server subnet, the SSH server (port 22) must be accessible from outside. It is configured on R2 with the corresponding Destination NAT (DNAT) entries (also known as static NAT). Assuming that the firewall access control list (ACL) is correctly defined on the R2 interfaces, the rules to be added to the R2e2OUT interface are: (source prefix, source port, destination prefix, destination port, protocol, Accept)

13.	A	any,>=1024,ssh,22,TCP,A	B	ssh,22,any,any,TCP,A	C	any,any,any,22,UDP,A	D	ssh,22,any,>=1024,TCP,A

The ACL defined on R3 to support the defined configuration must include:

14.	R3ppp0IN	A	any, any, any, any, IPinIP, A	B	any, - ,R3ppp0, - , IPinIP, A	C	any, any, P+T, any, any, A	D	R2ppp0,-, R3ppp0, - , IPinIP, A

Device P1 (plant subnet) has an empty ARP table and empty DNS cache. It runs ping www.fib.upc.edu.

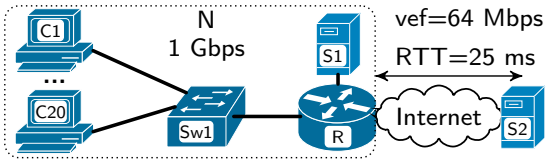
How many ARP messages does P1 send and how many IP datagrams enter and exit through R3ppp0 until the ICMP ECHO Reply message arrives?

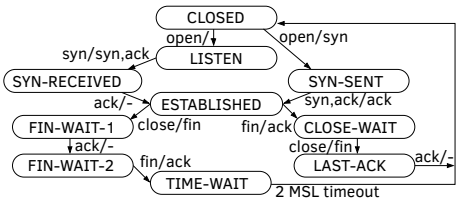
15.	A	1, 1, 1	B	1, 2, 2	C	2, 2, 1	D	2, 2, 2

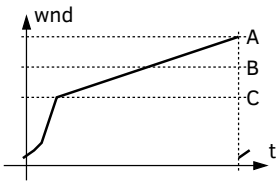
Examen final de Xarxes de Computadors (XC)	Grau en Ingeniería Informàtica	08/06/2026	Primavera 2026

Duration: 1h30m. All questions have a single true answer. Mark the answers on the back.

Problem TCP (2,5 points)







In the network N in the figure there are 20 PCs (C1$\cdots$C20) and a server S1. All links in the network N are Full Duplex Gigabit Ethernet. The router R has a **1 MB** (MByte) buffer. Between the router and the Internet server S2 there is an average of 64 Mbps and an RTT=25 ms. All TCP connections use an advertised window **awnd=100 kB** (kByte).

The throughput (vef), transmission window (W), and average RTT of C1, when there is only C1 transmitting a file to S1 will be:

1. vef: A. 100 Mbps B. 50 Mbps C. 1 Gbps D. 800 kbps
2. W: A. 5 kB B. 50 kB C. 1,5 kB D. 100 kB
3. RTT: A. 0,04 ms B. 0,1 ms C. 0,8 ms D. 6,4 ms

When C1 finishes sending the file it closes the socket. Tell which flag is plausible to be sent in the last segment of data that C1 sends, and the state of C1's socket after sending that segment:

4. flag: A. SYN=1 B. RST=1 C. ACK=0 D. FIN=1
5. state: A. FIN-WAIT-1 B. FIN-WAIT-2 C. CLOSE-WAIT D. LAST-ACK

When S1 receives the previous segment it closes the socket. The state of the S1 socket will be after closing the socket will be:

6. state: A. LAST-ACK B. FIN-WAIT-2 C. FIN-WAIT-1 D. CLOSE-WAIT

The throughput (vef), transmission window (W) and average RTT of C1, when the 20 PCs C1-C20 transmit to S1 are:

7. vef: A. 500 Mbps B. 100 Mbps C. 1 Gbps D. 50 Mbps
8. W: A. 37,5 kB B. 50 kB C. 100 kB D. 25 kB
9. RTT: A. 18 ms B. 0,8 ms C. 100 ms D. 6 ms

What is approximately the value of the following parameters of C1, when there is only C1 sending a file to S2:

10. optimal window: A. 100 kB B. 25 kbps C. 200 kB D. 75 kB
11. throughput: A. 64 Mbps B. 32 Mbps C. 8 Mbps D. 1 Gbps
12. average transmission window: A. 5 kB B. 50 kB C. 1500 Bytes D. 100 kB
13. average RTT: A. 42 ms B. 31 ms C. 125 ms D. 25 ms
14. R's buffer occupancy: A. 25 kbps B. buit C. 100 kB D. 75 kB

The figure above shows the evolution of the transmission window (wnd) of C1, when the 20 PCs C1-C20 transmit to S2. The maximum window (A), average (B) and C value, in kB, are approximately:

15. A: A. 50 B. 100 C. 60 D. 25
16. B: A. 20 B. 37,5 C. 75 D. 45
17. C: A. 30 B. 37,5 C. 50 D. 25

Assuming MSS=1500 bytes, say what it is approximately the duration, measured in RTT, of the slow start (SS) and congestion avoidance (CA) phase in the previous figure:

18. SS: A. 5 B. 7 C. 10 D. 3
19. CA: A. 30 B. 20 C. 15 D. 25

Approximately how many segments will be sent during the slow start (SS) and congestion avoidance (CA) phase in the above figure:

20. SS: A. 60 B. 5 C. 35 D. 45
21. CA: A. 200 B. 600 C. 1200 D. 400

What is the throughput (vef) and average RTT of C1, in the above situation:

22. vef: A. 10 Mbps B. 3,2 Mbps C. 50 Mbps D. 64 Mbps
23. RTT: A. 112,5 ms B. 25 ms C. 200 ms D. 100 ms

What will happen to the above values if C1 uses awnd=200 kB, while the other PCs use awnd=100 kB:

24. A. C1 will have double RTT than the other PCs
    B. C1 will have double the vef than the other PCs
    C. it will increase the RTT of all PCs
    D. there will be no changes in the vef and RTT of C1

What will happen, approximately, when the 20 PCs C1-C20 transmit and the 10 PCs C11-C20 finish transmitting:

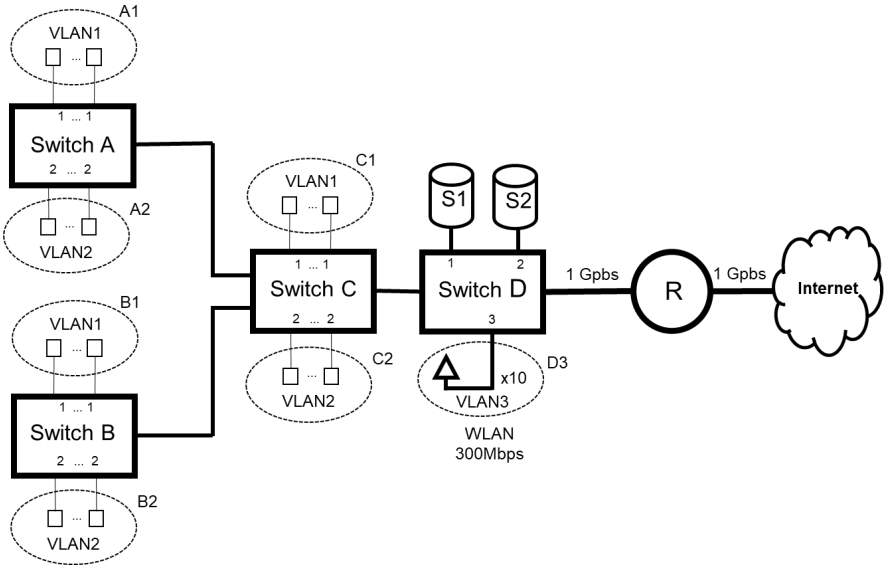
25. A. the transmission window of C1 will not change
    B. the RTT of C1 will be halved or more
    C. the transmission window of C1 will be reduced by half or more
    D. the transmission window of C1 will increase to double or more

Examen final. Xarxes de Computadors (XC), Grau en Enginyeria Informàtica	08/06/2026	Spring 2026	1

Total time: 2 hours 45 minutes. Single answer. All score p if correct, -p/3 if wrong, 0 if no answer. Clearly mark answers in pencil on the back.

Problem LAN (1.25 points; estimated time 20 minutes)

The figure shows the local network and its connection to the Internet. All Ethernet switch ports are Full Duplex at 100 Mbps, except those indicated at 1 Gbps. There are 3 VLANs. VLAN1 consists of 10 devices at A1, 5 devices at B1, 10 devices at C1, and server S1. VLAN2 has 10 devices at A2, 5 at B2, 10 at C2, and S2. VLAN3 has one Access Point and 10 mobile devices (D3). The Ethernet switches have the flow control mechanism enabled.



Only the VLAN1 clients (A1, B1, C1) are transmitting. They all transmit to a server on the Internet. The maximum effective speed they can achieve is:

1.	B1	A	1.66 Mbps	B	4 Mbps	C	0.83 Mbps	D	8.33 Mbps
2.	A1	A	4 Mbps	B	1.66 Mbps	C	8.33 Mbps	D	0.83 Mbps
3.	C1	A	8.33 Mbps	B	0.83 Mbps	C	1.66 Mbps	D	4 Mbps

Which switch ports transmit PAUSE frames in the previous case?

4.		A	SwD-C	B	SwC-A	C	SwD-S1	D	SwA-A1
5.		A	SwA-C	B	SwB-B1	C	SwC-B	D	SwB-C
6.		A	SwC-D	B	SwC-C1	C	R-SwD	D	cap

Only the VLAN1 clients are transmitting. They all transmit to server S2. The maximum effective speed they can achieve is:

7.	A1	A	0.83 Mbps	B	4 Mbps	C	1.66 Mbps	D	8.33 Mbps
8.	B1	A	8.33 Mbps	B	1.66 Mbps	C	4 Mbps	D	0.83 Mbps
9.	C1	A	4 Mbps	B	8.33 Mbps	C	0.83 Mbps	D	1.66 Mbps

Server S1 transmits to all VLAN1 clients. Indicate which port activates flow control and the reception speed of the clients.

10.		A	SwD-C, B1 a 4 Mbps	B	SwC-D, A1 a 1.66 Mbps	C	none, B1 a 4 Mbps	D	SwD-S1, C1 a 4 Mbps

Number of times the Ethernet frame generated by the indicated client with destination S1 is transmitted.

11.	A2	A	3	B	4	C	5	D	6
12.	B1	A	6	B	5	C	4	D	3
13.	B2	A	5	B	4	C	3	D	6
14.	C2	A	3	B	2	C	5	D	4

If there is activity between all devices in the three VLANs, S1 and S2, how many different MAC addresses will be in the switching tables of switches C and D?

15.	C, D	A	63, 63	B	25, 25	C	26, 36	D	35, 37

All VLAN1 clients transmit to S1 and all VLAN2 clients transmit to S2. The effective speed that each client can achieve is:

16.	B2	A	1.66 Mbps	B	5 Mbps	C	0.83 Mbps	D	0.454 Mbps
17.	C1	A	8.33 Mbps	B	2 Mbps	C	4.54 Mbps	D	9.08 Mbps
18.	A1	A	4 Mbps	B	0.227 Mbps	C	0.454 Mbps	D	2 Mbps

In the previous case, if the link between C and D is upgraded to 1 Gbps, the effective speed that each client can achieve is:

19.	A1	A	0.454 Mbps	B	2.72 Mbps	C	0.227 Mbps	D	2 Mbps
20.	B2	A	2 Mbps	B	4.54 Mbps	C	0.908 Mbps	D	0.454 Mbps
21.	C2	A	4.54 Mbps	B	0.454 Mbps	C	45.45 Mbps	D	9.08 Mbps

Only the D3 clients (WLAN) are transmitting; all towards S2. Assume the WLAN efficiency is 90% (10% of time lost due to CSMA/CA).
The access point is connected to a 100 Mbps port. The effective speed of each client is:

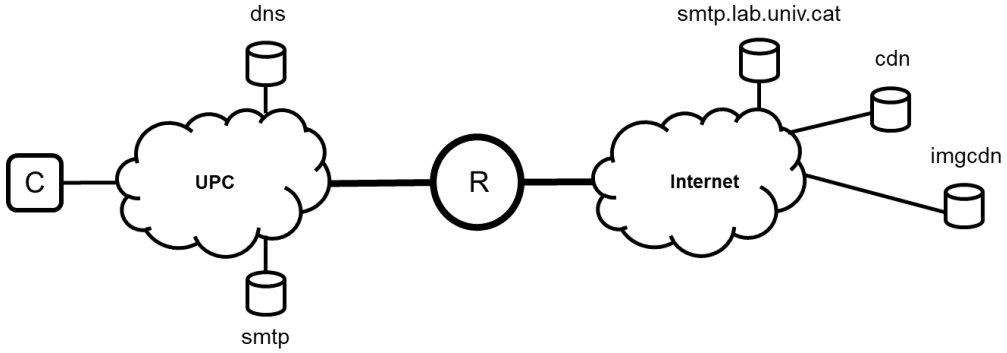
22.	D3	A	30 Mbps	B	9 Mbps	C	10 Mbps	D	100 Mbps

Examen final. Xarxes de Computadors (XC), Grau en Enginyeria Informàtica	08/06/2026	Spring 2026	1

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Problem APP (1.25 points; estimated time 20 minutes)

The figure shows a UPC network client C, the local DNS and email (SMTP) servers, and its connection to the Internet with some external servers.



The DNS server cache is empty. The RTT within the UPC network is 10ms. The client sends an email to user@lab.univ.cat. The RTT from UPC to the remote mail server is 150ms. Once the SMTP connection is established (the client sends HELO and receives the response), the message transmission takes 500ms (including FROM, RCPT, DATA through to QUIT). The SMTP protocol disconnects once the message transmission is complete.

The estimated time to transmit the message from client C to the local UPC server is:

1. A 510ms B 530ms C 520ms D 500ms

The UPC SMTP server sends the email to the recipient. For simplicity, the RTT to each domain authority is 100ms.

Indicate the resolution time of the MX record and the total time to transmit the message to the recipient's mailbox (in ms).

2. A 400, 900 B 10, 800 C 410, 1210 D 210,1010

Next, the UPC SMTP server sends the same message to copy@lab.univ.cat.

Indicate the DNS resolution time and the transmission time to the recipient's mailbox.

3. A 10, 650 B 0, 800 C 10, 800 D 100, 650

The message contains a few words and a PDF file (binary file) and the effective transmission speed is 100Mbps. With this data, if the transfer takes 500ms the total size of the transmitted message is less than 6250KB. The Content-Transfer-Encoding field and the maximum size of the attached file are:

4. A base64, 4687'5KB B base64, 3750KB C quoted-printable, 4680KB D quoted-printable, 3750KB

In the header of this email, the value of the Content-Type field can be:

5. A multipart/mixed B application/png C UTF-8 D quoted-printable

The client connects to the CDN HTTP server. The RTT between C and CDN is 300ms. The client downloads the index.html page (fits in a single segment) and then downloads the remaining objects sequentially. The page contains: a header (one segment), 5 icons (one segment each), and an image hosted on an external server IMGCDN with an RTT of 200ms. The time to download the image after the connection has been established and the GET has been received is 600ms. The local DNS server has the IP address resolution for CDN and IMGCDN. The browser client cache is empty. The client only establishes a single TCP connection per HTTP server and uses HTTP 1.0 (persistent without pipelining).

Number of DNS resolutions, TCP connections, and HTTP GET commands needed to download the complete page.

6. A 0, 2, 3 B 2, 2, 3 C 2, 1, 8 D 2, 2, 8

The total download time of the complete page (in ms) is approximately:

7. A 3420 B 2320 C 3400 D 3220

The client can establish as many simultaneous TCP connections as needed, and also simultaneous with DNS queries.

The download time of the index.html file only (in ms) is:

8. A 310 B 610 C 300 D 600

Once the index.html file has been downloaded, it can initiate the connections in parallel:

9. A 2 tcp, 1 http get B 1 dns, 1 tcp, 1 http get C 1 dns, 5 tcp, 1 http get D 6 tcp, 1 http get

Taking into account the operations that can be performed simultaneously, the time to download the header (in ms) is:

10. A 600 B 310 C 610 D 300

Taking into account the operations that can be performed simultaneously, the time to download the image (in ms) is:

11. A 1010 B 800 C 1000 D 600

Taking into account the operations that can be performed simultaneously, the time to download the five icons in parallel (in ms) is:

12. A 1500 B 600 C 3000 D 300

Taking into account the operations that can be performed simultaneously, the time to download the complete page (in ms) is:

13. A 1600 B 1210 C 910 D 1620

The client accesses the page again after a few seconds and the downloaded objects have not been modified. HTTP uses the keep-alive connection option.

The page reload time (in ms) will be:

14. A 300 B 210 C 600 D 10

The command used is:

15. A POST B GET C HEAD D UPDATE